

Bhopal CivicConnect — फ़ेज़ 15 सारांश (Notifications)

Ab app me in-app notifications hain (fetch-on-load, jaisa plan me decide hua tha — real-time Socket.io abhi bhi stretch goal hai)। Citizen ko notification milta hai jab uski complaint assign/status-update hoti hai; department ko notification milta hai jab unhe naya complaint assign hota hai ya citizen resolution confirm/reject karta hai। Poora flow test hokar verified (2026-09-10)।

बैकएंड — server/src/

	क्या काम करती है
models/Notification.js	Naya model — user (ref), complaint (ref, optional), type (assignment / status_update / verification_confirmed / verification_rejected), message, read (Boolean, default false)। { user, createdAt } par index — hamesha ek user ki notifications newest-first fetch hoti hain।
services/notification.service.js	Naya service — notifyUser() ek user ke liye notification banata hai, notifyDepartment() us department ke sabhi department_user accounts dhoondhkar ek saath insertMany se notify karta hai। Dono try/catch me non-blocking hain (Phase 6 ke reverseGeocode wale pattern jaisa) — notification fail hone par bhi complaint ka actual assign/status-update/verify action kabhi block nahi hota।
controllers/notification.controller.js	getMyNotifications (latest 30, newest-first, complaint title/status ke saath populated) aur markNotificationRead (ownership check controller ke andar, phir findByIdAndUpdate se read: true)।
routes/notification.routes.js	Sirf 2 routes — GET / aur PATCH /:id/read — bilkul PROJECT_PLAN.md me documented API shape jitna hi, koi extra endpoint (jaise "mark all read") add nahi kiya।
app.js	notificationRoutes ko /api/notifications par mount kiya — analyticsRoutes ke baad, notFound/errorHandler se upar।
controllers/complaint.controller.js	3 existing functions me notify calls hook kiye: assignDepartment (citizen + department dono ko), updateComplaintStatus (citizen ko), verifyComplaintResolution (department ko, sirf jab complaint kisi department se assigned ho)।

फ्रंटएंड — client/src/

	क्या काम करती है
services/notifications.js	2 simple wrapper functions — getNotifications(), markNotificationRead(id) — matching backend routes ko call karte hain।
components/NotificationBell.jsx	Naya component — Navbar me bell icon, unread count ka red badge, click par dropdown panel khulta hai (notifications list, unread ko halka teal background se highlight kiya, type ke hisaab se color-coded dot)। Ek notification par click karne se wo read mark ho jaata hai aur agar us notification se koi complaint juda ho to uski detail page par navigate ho jaata hai। Panel ke bahar click karne se apne aap band ho jaata hai।
components/Navbar.jsx	NotificationBell ko "My Complaints" link aur profile icon ke beech add kiya — sirf logged-in users ko dikhta hai।

क्यों ऐसे किया (viva के लिए)

Sirf 2 endpoints (GET /, PATCH /:id/read) kyun banaye, "mark all read" jaisa convenience endpoint kyun nahi — PROJECT_PLAN.md me API section me Notifications ke liye bilkul yahi 2 routes documented the। Scope ko wahi rakha jo plan me approved tha, taaki phase apne documented shape se match kare aur feature-creep na ho।

notifyUser/notifyDepartment ko try/catch me non-blocking kyun rakha, error throw kyun nahi kiya — agar notification create hote waqt DB me koi issue aaye, to us wajah se citizen ki complaint assign hone ya status update hone jaisi actual, zaroori action fail nahi honi chahiye। Ye bilkul wahi non-blocking pattern hai jo Phase 6 ke reverseGeocode me use hua tha — ek "nice to have" side-effect ki failure kabhi bhi main operation ko rokni nahi chahiye।

markNotificationRead me findByIdAndUpdate + controller-level ownership check kyun kiya, route-level authorize() kyun nahi — ye check role-based nahi hai ("kya user admin hai"), balki per-document hai ("kya ye wahi user hai jiske liye ye notification bani thi") — bilkul wahi pattern jo Phase 13 ke verifyComplaintResolution me use hua tha, wahi dobara reuse kiya।

notifyDepartment poore department ke saare department_user accounts ko kyun notify karta hai, kisi ek specific assignee ko kyun nahi — app me complaint assignment department-level hai (Phase 9 se), koi individual staff-member "assignee" concept nahi hai। Isliye jab bhi department ko kuch batana ho, uss department ke saare department_user accounts ko ek saath (insertMany se, ek-ek karke Notification.create() call kiye bina) notify karna hi sahi tha।

Challenges / मल्टी errors aur unka resolution

Is phase me koi naya bug nahi mila: backend notification-triggers (assign, status-update, verify confirm/reject), aur frontend bell (badge count, dropdown, read-on-click, navigate) pehli hi testing pass me sahi kaam kar gaye। Populate hone ke baad field access (jaise complaint.reporter._id) aur ownership-check pattern dono pehle se sikhe hue lessons the (Phase 10, Phase 13), isliye is baar shuru se hi sahi likhe gaye।

Aage dhyan rakhne wali baat

Notifications abhi sirf fetch-on-load hain — agar user tab khula chhod kar baitha hai aur iske beech me koi naya notification bana, to wo tab reload/navigate kiye bina dikhega nahi (real-time push, jaise Socket.io, plan me explicitly stretch goal hai, MVP me nahi hai)। Saath hi, list sirf latest 30 notifications tak limited hai, bahut purani notifications automatically hat jaati hain (koi pagination/archive nahi), jo is project ke scale ke liye theek hai.

टेस्टिंग स्टेटस

Admin assign karta hai -> citizen + department dono ko notification -> pass | Department status update karta hai -> citizen ko notification -> pass | Citizen confirm karta hai -> department ko notification -> pass | Citizen note ke saath reject/reopen karta hai -> department ko note ke sath notification -> pass | Notification par click -> read mark + complaint detail par navigate -> pass | Page reload ke baad read-status persist -> pass | Zero notifications wale account me empty state -> pass | Logged-out state me bell hidden -> pass

Git checkpoint: b4fefb9 — "feat: Phase 15 - notifications (in-app, fetch-on-load)" — master branch par committed (eeb9456 Phase 14, 1986f69 Phase 13 ke baad)। Koi naya secret ya .env change is phase me nahi hua।